An evolutionary adapter for LeWM

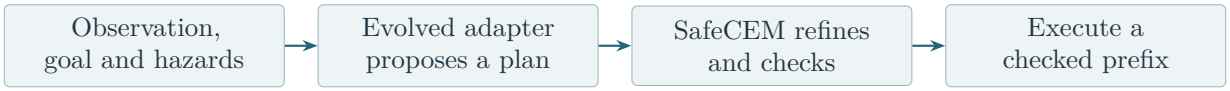
A small controller that learns to propose useful behaviour

Starting point. LeWM predicts task progress from candidate actions. A separate learned motion model and supplied geometry predict clearance from newly introduced hazards. SafeCEM searches for a useful checked plan. The proposal is to add a small learned controller that supplies better starting action sequences.

What evolution would learn

Each individual is a set of adapter-network weights. Given the current observation, goal and hazard information, the network proposes a short action sequence or a bounded correction to a proposal. Evolution evaluates candidate networks across development episodes and improves their weights. The resulting adapter can be reused in different situations, with its weights fixed during evaluation.

LeWM and the motion predictor remain frozen. SafeCEM refines the proposal using its existing task score and safety checks. This makes evolution responsible for learning a behavioural proposal, while preserving the current planning interface.



Training and execution

1. **Provide causal inputs.** Use world-model features, goal information, native proprioception and explicitly supplied hazard geometry. Our virtual hazards are not automatically visible in LeWM's images.
2. **Evolve network weights.** CMA-ES is one candidate optimizer. Evaluate each network with the same frozen models and planner. Rank primarily by actual safe task completion, with secondary criteria specified before the experiment.
3. **Refine and check each proposal.** SafeCEM also retains ordinary proposals so search can recover from a poor adapter. Keep the safety margin and prediction allowance fixed. Execute only a checked prefix; refuse if none exists.
4. **Freeze before testing.** Select using development evidence, then evaluate on fresh source cases without updating adapter weights. Observe and replan after each executed prefix.

Why this might help

The adapter might learn to propose an initial sideways movement when the direct route is obstructed. SafeCEM could then refine it into a useful detour. A good proposal could improve completion or reduce wasted online search. A poor one could bias search towards an unhelpful route. These are hypotheses; poor initialization has not been established as the cause of our current stalls.

Research question. Can an evolution-trained proposal adapter improve safe completion around frozen LeWM under fixed safety requirements and matched budgets?

Current status. This behavioural proposal adapter has not been implemented or evaluated in our LeWM pipeline. The next page explains the supporting evidence and the comparisons needed to assess it.

What the literature establishes

Working precedents and the questions for our study

Four close precedents

- **POPLIN, ICLR 2020 [1].** A policy proposes action sequences for CEM refinement; another variant searches policy parameters. Persistent policies use cloning, adversarial training or averaging. Models update during training, and our hazard-safety setting is not tested.
- **World Models, NeurIPS 2018 [2].** CMA-ES evolves a compact controller over learned representations. The CarRacing controller has 867 parameters and trains in the actual simulator. A separate VizDoom experiment demonstrates learned-model training and transfer. Neither establishes our safe-completion result.
- **Behavior Policy Learning, 2022 [3].** CMA-ES fine-tunes an imitation-initialized neural target generator above fixed low-level control. One optimized policy completes 8/12 physical pick-and-place configurations. This is task completion, not an introduced-hazard safety rate.
- **Safe-LOOP, CoRL 2021 [4].** An actor guides population planning with learned dynamics and cost constraints. Actor learning uses gradients and models update during training. This supports constrained planning, not evolutionary adapter learning specifically.

What to take from these results

The components have worked, but the complete LeWM combination remains unverified in this focused search. That is not proof of novelty. Keep existing SafeCEM and learn a compact proposal. An imitation initialization is worth considering, with its data and training costs recorded.

POPLIN's parameter-space search can work with fixed zero base weights, so successful planning alone does not establish useful adapter learning.

Two separate questions to test

1. **Does the adapter help?** Compare the same planner alone, with an untrained adapter, and with the evolved adapter. If imitation initializes it, include the imitation-only adapter before evolutionary fine-tuning.
2. **Does evolution add value?** Compare evolution, parameter random search and ordinary learning on the same adapter representation, information and budgets. Use repeated searches and fresh evaluation source cases.

Evidence to retain. Report safe completion, violations, unfinished episodes and engagement with denominators. Count training and online work separately and in total. Predicted feasibility is not guaranteed safety. Training entirely in imagination needs closed-loop predictions of all inputs and termination, beyond our current open-loop evaluators.

Primary reading

[1] Wang and Ba. *Learning to Plan with Policy Networks.* ICLR 2020; preprint 2019.

[2] Ha and Schmidhuber. *Recurrent World Models Facilitate Policy Evolution.* NeurIPS 2018.

[3] Tsinganos et al. *Behavior policy learning.* Frontiers in Robotics and AI, 2022.

[4] Sikchi, Zhou and Held. *Learning Off-Policy with Online Planning.* CoRL 2021; proceedings 2022.